ERICK ALBERTO SÁNCHEZ ARANDA

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EDUCATION

Tecnológico de Monterrey (ITESM) **Aug. 2024 – Expected Jun. 2028**

B.S. in Computer Science and Technology | GPA: 9.0/10.0 *Guadalajara, Jalisco*

- **Coursework:** Data Structures and Algorithms, Advanced Algorithms, Computer Architecture, Computer Networks, Cybersecurity

EXPERIENCE

Undergraduate Research Assistant, Water Filtration Telemetry **Aug. 2026 – Present**

Tecnológico de Monterrey *github.com/uno21858/water-column-telemetry*

- Building the telemetry stack for a four-column water filtration rig: interrupt-driven ESP8266 firmware in C++ that counts flow-sensor pulses per column, a Supabase (PostgreSQL) schema for the readings, and a Next.js dashboard deployed on Vercel.
- Kept ISRs in IRAM, as the ESP8266 requires, and read shared pulse counters inside critical sections to avoid race conditions.

Undergraduate Research Assistant, Computer Vision **Feb. 2026 – Jun. 2026**

Tecnológico de Monterrey | Advisor: Prof. Atoany Fierro Radilla *github.com/uno21858/traffic-sign-cnn*

- Trained a custom ~628K-parameter CNN in PyTorch on GTSRB traffic signs (43 classes), reaching 92% on the official test set.
- Set up GPU training on a self-hosted RHEL server (Docker, NVIDIA Container Toolkit) and served the model with FastAPI.

Administrative Assistant (Part-time) **Aug. 2022 – Present**

Transportes Sánchez Barcelata *Guadalajara, Jalisco*

- Built FDM (Factura Diesel Manager) on own initiative: a Python desktop app (PySide6, Selenium) that downloads diesel invoices from Mexico's tax authority (SAT) and totals liters and cost, replacing a manual reconciliation workflow; ships as a Windows installer (PyInstaller + NSIS).
- Adapted FDM's architecture into a bulk payroll-invoice downloader (PDF + XML) with batch scheduling and retry logic to pull 5,000+ invoices for a full fiscal-year compliance audit within SAT's daily download limits.

LEADERSHIP & ACTIVITIES

PawnGuard, Cybersecurity Student Group **Oct. 2025 – Present**

Treasurer, previously Infrastructure Staff *Guadalajara, Jalisco*

- Co-organized Hack2Dawn, an internal jeopardy-style CTF sponsored by MLH, 330ohms and Google Cloud; engineered its CI/CD pipeline (GitHub Actions, Docker) that provisions CTFd on a VPS and monitored the server during the live event.
- Developed two OSINT challenges for the CTF: GitHub commit forensics and search-engine dorking for exposed documents.

PROJECTS

Cuentas Claras LATAM | *Flutter, Supabase, AWS* **2025 – Present**

Expense-splitting mobile app | solo developer *cuentasclaraslatam.com*

- Shipped to the App Store and Google Play with AWS Cognito federated sign-in (Google, Apple), a pre-token-generation Lambda that injects JWT claims for Supabase, RevenueCat subscriptions, and FCM push through AWS Lambda.
- Designed the data layer so the client never queries tables directly: reads and writes go through SECURITY DEFINER RPCs, ~170 in production, that check the caller against the Cognito JWT; RLS on all 21 tables, auto-enabled on new ones by an event trigger; bank details encrypted with pgcrypto and a key held in Supabase Vault.
- Ran authorized IDOR tests against the app's API: intercepted Flutter traffic with reFlutter, Frida/objection and Burp Suite and replayed requests with a second account's JWT; found and fixed an IDOR that let a regular user modify other users' data.

Home Water SCADA | *C++, ESP32-S3, LVGL, KiCad* **Sep. 2026 – Present**

Monitoring and chlorination controller for a home water system | self-directed *github.com/uno21858/home-water-scada*

- ESP32-S3 controller with a 7-inch touch HMI reading a hall-effect flow sensor, two pressure transducers via a 16-bit I2C ADC (ADS1115) and a 4–20 mA level transmitter, plus a MOSFET-driven 12 V mixing pump; meters water volume to compute chlorine dosing.
- Wrote the HMI in C++ with LVGL (process diagram, readings, manual controls) on a desktop simulator, then flashed with PlatformIO.
- Designed the sensor node in KiCad (single 12 V input with 5 V regulation and 24 V boost, fused and zener-clamped input, flyback diode), built it on perfboard and bench-validated every channel before integration; 3D-printed the display case and sensor mounts.

LLM Fine-Tuning | *PyTorch, PEFT, Ollama* **2026**

Self-directed | self-hosted RHEL server with two RTX 3060 GPUs

- Fine-tuned Qwen3-8B with QLoRA on 200K+ plain-text messages from personal WhatsApp and Instagram exports to model a conversational persona; merged adapters with PEFT, quantized to GGUF and served it locally via Ollama.

Zenalyse | *Raspberry Pi, AWS, Next.js* **Nov. 2025**

IoT environmental monitoring | 1st Place out of 8 engineering teams, Skilliket Symposium

- IoT system correlating sensor data (DHT11, MQ-135, PIR) with user stress levels: Python polling on a Raspberry Pi, AWS EC2 backend with PostgreSQL, Next.js dashboard.

SKILLS

Programming: Python, C++, Dart, SQL (PL/pgSQL), TypeScript/JavaScript, Java, CUDA, Bash
Embedded: ESP32 / ESP8266 (PlatformIO), LVGL, KiCad, I2C, ADC and 4–20 mA sensor interfacing, 3D printing (FDM)
ML & Data: PyTorch, PEFT (QLoRA), NumPy, Pandas, scikit-learn
Backend & Cloud: Supabase, PostgreSQL, AWS (Cognito, Lambda, EC2), Firebase, FastAPI
Frameworks & Tools: Flutter, Next.js, PySide6, Selenium, Docker, Git, GitHub Actions, Linux (RHEL, Ubuntu), Cloudflare (Zero Trust, Tunnels)
Security: Burp Suite, Frida, objection, reFlutter, Nmap, Wireshark
Certifications: AWS Certified Cloud Practitioner (Aug. 2025), Oracle Java Foundations (2023)
Languages: Spanish (native), English (professional working proficiency, Cambridge B2 First)